Computer Science

Exercises

EXERCISE: DESCRIBE THE FOLLOWING FUNCTIONS

Function A:

```
def circle_area(radius):
  return 3.14 * (radius**2)
```

Function B:

```
def list_summation(num_list):
  sum = 0

  for num in num_list:
    sum += num

  return sum
```

Function C:

```
def smallest_in_list(num_list):
  smallest = None

  for num in num_list:
    if smallest == None:
      smallest = num
    elif num < smallest:
      smallest = num

  return smallest
```

Function D:

```
def solve_quadratic(a, b, c):
  delta = (b**2) - (4 * a * c)

  if delta >= 0:
    x_1 = (-b - delta**0.5) / (2 * a)
    x_2 = (-b + delta**0.5) / (2 * a)
    return [x_1, x_2]
  else:
    return "No real solutions"
```

Function E:

```
def vector(x_a, y_a, x_b, y_b):
  x_ab = x_b - x_a
  y_ab = y_b - y_a

  return [x_ab, y_ab]
```

1. Describe the functionality of each of these functions

SOLUTION:

1.

A)

- This function is called `circle_area`, and takes in a parameter `radius`.
- It uses the formula $A = \pi r^2$ to calculate the circle's area.
- Then it returns the area

B

- This function is called `list_summation`, and takes in a parameter `num_list`
- It then makes a variable that will hold the sum of all numbers in the list
- It then loops over each number in the list, adding it to that holder variable
- And finally, returns the sum

C:

- This function is called `smallest_in_list`, and accepts a parameter `num_list`
- It then creates a variable for the smallest number
- It then loops over each number in the list
  - If there is no smallest number yet, then the smallest number is the current number
  - If there is a smallest number, and it's bigger than the current number, than the current number is the new smallest number
- Then it returns the smallest number

D:

- This function is called `solve_quadratic` and accepts three parameters `a`, `b` and `c`.
- It then uses the formula $\Delta = b^2 - 4ac$ and holds the delta in a variable
  - If $\Delta \geq 0$, then it calculates the two (or double) roots of the equation and returns them in a list.
  - Else ($\Delta < 0$), it will return the string "No real solutions"

E:

- This function is called `vector` and takes in 4 parameters: `x_a`, `y_b`, `x_b`, `y_a`
- It will then calculate the difference between `x_a` and `x_b` and hold it in a variable
- It will also calculate the difference between `y_a` and `y_b` and hold it in a variable
- Finally, it returns the two variables in a list